Software Requirements Specification (SRS)

for

Smart City Traffic Management System (SCTMS)

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Software Requirements Specification

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the requirements for the Smart City Traffic Management System (SCTMS). This document serves as the foundation for design, development, testing, and deployment. It defines the functional capabilities, external interfaces, performance benchmarks, and security constraints necessary to deliver a reliable, real-time traffic optimization solution.

1.2 Document Conventions

• Must: Mandatory requirement essential for operation.

• Should: Recommended requirement that adds value but is not critical.

• May: Optional requirement.

• Shall: Binding contractual requirement.

• R#: Unique requirement identifier for traceability.

1.3 Intended Audience

• Development Team — implement the system.

• Project Managers — plan timelines, resources, and scope.

• Quality Assurance Team — create test plans and validation protocols.

• City Transportation Department — validate urban mobility objectives.

• Emergency Services — ensure emergency vehicle prioritization.

• System Administrators — understand deployment and maintenance.

1.4 Product Scope

The SCTMS is a cloud-based, AI-driven platform designed to monitor and control traffic flow across an entire metropolitan city. It will monitor traffic through roadside cameras, inductive loop sensors, radar detectors, and public-transit GPS; dynamically optimize traffic signals; prioritize emergency vehicles; provide public traffic information; and generate analytics and reports.

The system aims to reduce average commute times by 25%, decrease emergency vehicle response times by 40%, and lower city-wide carbon emissions by 15% through reduced idling.

1.5 References

• IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
• ISO 14817: Intelligent Transport Systems (ITS) — Data Dictionary.
• NTCIP Standards.
• City of [Metropolis] Urban Development Plan 2025.
• System Business Requirements Document (BRD) v1.0.

2. Overall Description

2.1 User Needs

User Type	Needs Description
Traffic Control Center Operators	Centralized command dashboard for city-wide traffic, live cameras, emergency signal overrides, and alerts.
City Transportation Planners	Historical traffic data, congestion heatmaps, and intersection reports for infrastructure planning.
Emergency Vehicle Dispatchers	Green-wave route status and confirmation that intersections have been cleared.
Field Maintenance Technicians	Mobile alerts for malfunctioning signals, cameras, and sensors and maintenance updates.
General Public	Public app and VMS information showing traffic conditions, travel times, and alternate routes.
System Administrators	User management, signal parameters, system health, and third-party integrations.

2.2 Assumptions and Dependencies

2.2.1 Assumptions

- A-01: Intersections use programmable NTCIP-compliant traffic signal controllers.
- A-02: Cameras and sensors at major intersections stream continuously.
- A-03: High-bandwidth fiber connects field devices to the cloud.
- A-04: Emergency vehicles have real-time GPS tracking.
- A-05: Field devices have UPS-backed power.

2.2.2 Dependencies

- D-01: Google Maps API and OpenStreetMap for maps, routing, and geocoding.
- D-02: AWS Cloud Platform (EC2, RDS, S3, Lambda, Kinesis).
- D-03: CAD integration depends on API credentials and data mapping.
- D-04: 5G network for emergency and public-transit GPS transmission.

3. System Features and Requirements

3.1 Functional Requirements

3.1.1 User Authentication & Access Control

- R-101: Unique username/password, minimum 12 characters with special characters.
- R-102: RBAC roles: Operator, Planner, Dispatcher, Maintenance Technician, Admin.
- R-103: MFA via OTP for Admin and Operator accounts.
- R-104: Log all user actions with timestamps.

3.1.2 Real-Time Traffic Monitoring & Data Ingestion

- R-201: Ingest loop/radar data every 2 seconds including count, speed, and occupancy.
- R-202: Process IP-camera feeds with AI for density, accidents, and vehicle counts.
- R-203: Ingest bus and emergency GPS every 5 seconds.
- R-204: Calculate Traffic Congestion Index 0–10 for every road segment every 30 seconds.
- R-205: Alert Maintenance if a sensor is silent for more than 5 minutes.

3.1.3 Dynamic Traffic Signal Optimization (AI Engine)

- R-301: Use reinforcement learning to adjust signal timing from real-time congestion.
- R-302: Synchronize arterial signals to create Green Waves.
- R-303: Include time-of-day and special-event patterns.
- R-304: Allow manual Operator overrides with mandatory reason codes.
- R-305: Revert to a Fallback Timing Plan if the AI engine or cloud connection fails.

3.1.4 Emergency Vehicle Preemption (Green Wave)

- R-401: Detect emergency vehicles using GPS position and heading.
- R-402: Within 500 meters, preemptively turn the vehicle direction green and conflicts red.
- R-403: Clear preemption automatically after the vehicle passes.
- R-404: Show Green Wave Confirmation on the Dispatcher dashboard.
- R-405: Log preemption timestamp, vehicle ID, route, and duration.

3.1.5 Public Information & Communication

- R-501: Feed congestion, accident, and closure data to public VMS.
- R-502: Provide a public REST API for anonymized traffic data.
- R-503: Send public push notifications for major incidents.
- R-504: Provide estimated travel times for major routes.

3.1.6 Reporting & Analytics

- R-601: Generate daily corridor Traffic Performance Reports.
- R-602: Generate monthly reports ranking the 10 most congested intersections.
- R-603: Provide hourly, daily, and weekly congestion heatmaps.
- R-604: Report average emergency-response time saved through preemption.

3.2 External Interface Requirements

3.2.1 Hardware Interfaces

- HI-01: NTCIP signal controllers via SNMP over Ethernet.
- HI-02: ONVIF IP cameras receiving RTSP video streams.
- HI-03: Inductive loop and radar sensors via Modbus.

3.2.2 Software Interfaces (APIs)

- SI-01: CAD integration via REST API for emergency assignment data.
- SI-02: GPS integration via MQTT JSON payloads.
- SI-03: Google Maps API and OpenStreetMap for routing, geocoding, and rendering.
- SI-04: VMS integration via TCP/IP for dynamic messages.

3.2.3 User Interfaces (UI)

- UI-01: City-wide map with Green, Yellow, Orange, and Red congestion states.
- UI-02: Up to four simultaneous live camera feeds.
- UI-03: Signal Control Panel for manual timing changes.
- UI-04: Maintenance alert list sorted by severity/timestamp with Acknowledge and Assign actions.

3.3 Non-Functional Requirements

3.3.1 Performance & Scalability

- NFR-01: Process at least 10,000 sensors and cameras simultaneously.
- NFR-02: Compute timings for at least 1,000 intersections within 15 seconds.
- NFR-03: Load city traffic map in under 3 seconds.
- NFR-04: Trigger emergency preemption and communicate signal changes within 2 seconds.
- NFR-05: Support at least 500 concurrent dashboard sessions.

3.3.2 Reliability & Availability

- NFR-06: 99.99% uptime for critical signal-control and emergency-preemption functions.
- NFR-07: AWS Availability Zone failover within 60 seconds.
- NFR-08: Cloud outage causes controllers to revert to local Fallback Timing Plans.
- NFR-09: PostgreSQL backups every 6 hours in a separate AWS region.

3.3.3 Security & Compliance

- NFR-10: TLS 1.3 for field-device-to-cloud traffic.
- NFR-11: OAuth 2.0 for API access with 1-hour token expiration.
- NFR-12: Retain detailed audit logs for at least 5 years.
- NFR-13: Follow NIST Cybersecurity Framework guidelines.

3.3.4 Usability

- NFR-14: High-contrast colors and large icons for rapid decisions.
- NFR-15: Confirmation dialogs for critical actions.
- NFR-16: Contextual tooltips and help documentation.
- NFR-17: English, Spanish, and Mandarin interfaces.

3.3.5 Maintainability

• NFR-18: Docker containers and Kubernetes orchestration.

• NFR-19: Minimum 85% unit-test coverage and hardware-protocol integration tests.

• NFR-20: JSON logs aggregated to centralized ELK.

4. Appendixes

Appendix A: Data Dictionary

Data Element	Data Type	Description	Constraints
Intersection_ID	VARCHAR(50)	Unique traffic intersection identifier.	Primary Key, Mandatory.
Traffic_Signal_Controller_IP	VARCHAR(15)	IP address of NTCIP controller.	Valid IPv4 address.
Vehicle_Count	INTEGER	Vehicles detected in a time window.	Must be >= 0.
Average_Speed	DECIMAL(5,2)	Average road-segment speed (km/h).	0 to 120.
Congestion_Index	DECIMAL(2,1)	Congestion score, 0 Free Flow to 10 Gridlock.	0.0 to 10.0.
Emergency_Vehicle_ID	VARCHAR(20)	Unique emergency vehicle identifier.	Must match CAD records.
Preemption_Status	ENUM	Emergency preemption status.	ACTIVE, CLEARED, PENDING.
Signal_Timing_Plan	JSON	Green/yellow/red timing structure.	Example: green 30, yellow 4, red 26.
Manual_Override_Reason	VARCHAR(255)	Reason for manual signal override.	Mandatory when Manual_Override = TRUE.

Appendix B: Acronyms and Abbreviations

Acronym	Full Form
SCTMS	Smart City Traffic Management System
ITS	Intelligent Transport Systems
NTCIP	National Transportation Communications for ITS Protocol
VMS	Variable Message Signs
CAD	Computer-Aided Dispatch
GPS	Global Positioning System
RTSP	Real-Time Streaming Protocol
ONVIF	Open Network Video Interface Forum
MQTT	Message Queuing Telemetry Transport
SNMP	Simple Network Management Protocol
AI	Artificial Intelligence
ML	Machine Learning
SLA	Service Level Agreement
K8s	Kubernetes
ELK	Elasticsearch, Logstash, Kibana
TPS	Transactions Per Second

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